

EXPERIMENT – 18

CODE:

```
#include <reg51.h>

sbit PWM = P1^0;

unsigned char count = 0;
unsigned char duty = 175; // Change value from 0–200

void timer0_ISR(void) interrupt 1
{
    TH0 = 0xFC;          // Reload for 1 ms delay
    TL0 = 0x66;

    count++;

    if(count < duty)
        PWM = 1;          // ON time
    else
        PWM = 0;          // OFF time

    if(count >= 200)      // Total period = 200 ms
        count = 0;
}

void main(void)
```

```

{
    TMOD = 0x01;          // Timer 0, Mode 1 (16-bit)

    TH0 = 0xFC;           // 1 ms preload
    TL0 = 0x66;

    IE = 0x82;            // EA = 1, ET0 = 1

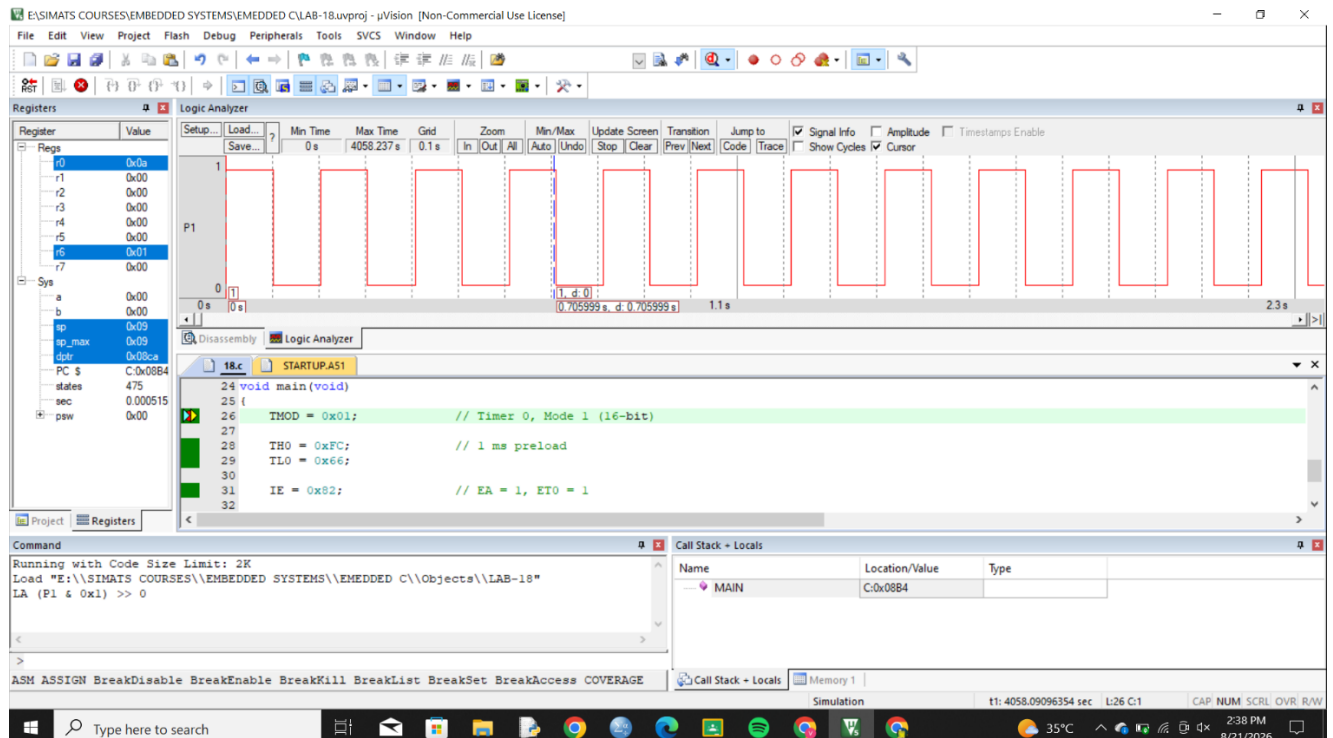
    TR0 = 1;              // Start Timer 0

    while(1);
}

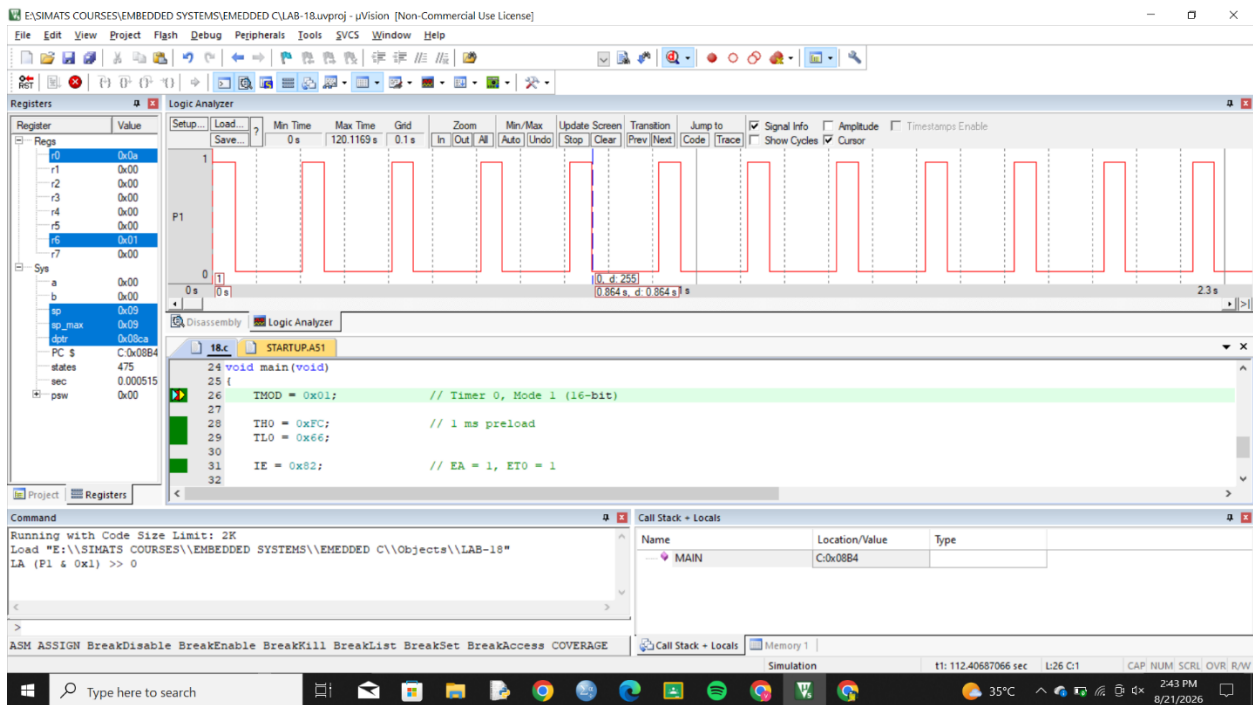
```

OUTPUT:

50% duty cycle:



25% Cycle Duty



75% Cycle Duty:

